

Dr. Doc:

Physiology Note

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HUMAN RESPIRATORY PHYSIOLOGY

1 STRUCTURAL ANATOMY & MECHANICS OF VENTILATION

Overview

The respiratory system is the strategic interface required to maintain cellular homeostasis through the precisely regulated exchange of gases. This physiological necessity is achieved through four distinct stages:

 Pulmonary ventilation — the mechanical inflow and outflow of atmospheric air.

 External respiration — the exchange of O_2 and CO_2 between the alveoli and pulmonary capillaries.

 Transport — the systemic movement of these gases via the blood.

 Internal respiration — the final exchange between systemic capillaries and the tissues to fuel cellular metabolism.

By ensuring the continuous delivery of oxygen and the removal of metabolic carbon dioxide, the system prevents cellular toxicity and maintains the energetic integrity of the organism. ⚡

💡 High-yield:

The overall pathway is:

Atmosphere → Alveoli → Blood → Tissues → Blood → Alveoli → Atmosphere ↻

◆ Functional Taxonomy

Beyond the primary gas exchange, the respiratory system performs a wide array of vital non-respiratory functions:

👤 Respiratory Functions

*Gas Conduction: Filtering, warming, and humidifying inspired air.

*Gas Exchange: The absorption of O_2 and excretion of CO_2 across the respiratory membrane.

🛡️ Non-Respiratory Functions

⚖️ Acid-Base Balance: Rapidly regulating blood pH by modulating CO_2 levels.

🗣️ Vocalization: Phonation through the controlled movement of air across the vocal cords in the larynx.

💉 Metabolic Activation: The conversion of Angiotensin I to the active vasoconstrictor Angiotensin II.

🛡️ Defense Mechanisms: Utilizing the "mucus escalator," ciliary beating, and the phagocytic activity of alveolar macrophages.

Hemodynamic Roles: Serving as a blood reservoir and enhancing venous return to the heart.

💡 Remember:

The lungs are not simply "gas-exchange organs." They also contribute significantly to pH regulation, defense, metabolism, speech, and circulation.

◆ Anatomical vs. Physiological Divisions

🧑 Anatomically

↑ Upper Respiratory Tract: Includes the nasal cavity and the pharynx.

↓ Lower Respiratory Tract: Begins at the larynx and continues through the trachea, bronchi, and lungs.

🫁 Physiologically

We classify the system into functional "generations" based on branching:

👉 Conducting Zone — Generations 0–16

*Trachea: Generation 0

*Bronchi: Generations 1–11

*Terminal bronchioles: Generations 12–16

✳️ Notably, conductive structures of a size down to approximately 1 mm are termed bronchi.

➡️ This zone is dedicated to air conditioning; no gas exchange occurs here.

🫁 Respiratory Zone — Generations 17–23

*Respiratory bronchioles: 17–20

*Alveolar ducts: 21–23

*Alveolar sacs

➡️ These are the exclusive sites of gas exchange.

💡 High-yield distinction:

Conducting zone = moves and conditions air 🗣️

Respiratory zone = exchanges gases 🔄

🛡️ The Mucociliary Defense

The "mucus escalator" is a critical defense mechanism where hair-like cilia line the passages down to the bronchioles.

◆ These cilia move particulate matter and microbes trapped in mucus toward the pharynx for elimination.

🚬 Ciliary motility is exceptionally sensitive to noxious agents; smoking a single cigarette can cause ciliary paralysis for several hours.

This paralysis, coupled with nicotine-induced hypersecretion of mucus, leads to the accumulation of secretions, which causes airway obstruction and invites chronic infection. 🦠

💡 Simple sequence:

Particles → mucus traps them → cilia move mucus upward → pharynx → elimination ↑

⚙️ The Physics of Airflow

Pulmonary ventilation utilizes Boyle's Law:

📐 $V \propto 1/P$

Changes in thoracic volume drive the pressure gradients necessary for flow.

💨 Quiet Inspiration

- Active process
- Contraction of the diaphragm (the primary inspiratory muscle) and external intercostals
- Increases thoracic volume
- Creates sub-atmospheric pressure
- Air moves inward ←

😴 Quiet Expiration

- Primarily a passive process

- Driven by the elastic recoil of the lungs and chest wall

Forced Expiration

An active process requiring:

* Abdominal muscles:

Rectus abdominis

Obliques

* Internal intercostals

➡ These forcefully decrease thoracic volume.

 Important:

Inspiration = usually active 

Quiet expiration = usually passive 

Forced expiration = active 

The diaphragm is the primary muscle of inspiration and is innervated by the phrenic nerve (C3–C5). 

➡ 

Pressure Dynamics

During inspiration:

Intrapleural pressure becomes more negative.

Changes from approximately $-5 \text{ cm H}_2\text{O} \rightarrow -7.5 \text{ cm H}_2\text{O}$

This negative pressure expands the lungs.

Alveolar pressure drops to approximately $-1 \text{ cm H}_2\text{O}$.

This sub-atmospheric pressure draws air inward. 

During expiration:

Elastic recoil causes alveolar pressure to become supra-atmospheric ($+1 \text{ cm H}_2\text{O}$).

Air flows outward. 

➡ These mechanical pressures allow the transition of bulk air flow into the microscopic gas-exchange units.

Additional Pressure Concepts

◆ Intrapleural Pressure

This is the pressure within the pleural space surrounding the lungs.

It is normally negative relative to atmospheric pressure.

➡ This negative pressure helps keep the lungs expanded against the chest wall.

◆ Alveolar Pressure

This is the pressure inside the alveoli.

Inspiration: slightly negative → air enters.

Expiration: slightly positive → air leaves.

No airflow: alveolar pressure equals atmospheric pressure.

◆ Transpulmonary Pressure

$$P_{TP} = P^A - P_{PL}$$

Where:

P^A = alveolar pressure

P_{PL} = intrapleural pressure

💡 It is the distending pressure of the lung.

➡ The greater the transpulmonary pressure, the more the lung tends to expand.

 Airway Resistance

Air must overcome resistance as it travels through the respiratory tract.

According to Poiseuille's relationship:

$$\text{Resistance} \propto 1/r^4$$

This means that a small decrease in airway radius can produce a large increase in resistance.

💡 Clinical connection:

This is why bronchoconstriction in asthma can dramatically increase the work required to breathe. 😞

2 ALVEOLAR DYNAMICS, PULMONARY COMPLIANCE & SPIROMETRY

The architecture of approximately 300 million alveoli is designed to maximize surface area while minimizing the metabolic "work of breathing."

⚡ Normally, the energy required for ventilation accounts for only 3–5% of total body expenditure.

💡 Main idea:

The alveoli provide a huge surface area for gas exchange while surfactant and lung elasticity help keep the work of breathing low.

🌐 Alveolar Cellular Architecture

The alveolar walls are composed of three specialized pneumocytes:

◆ Type I Pneumocytes

- Extremely thin squamous cells.
- Form the major lining.
- Facilitate rapid gas exchange. 🔄

💡 Their extreme thinness minimizes the distance gases must cross.

◆ Type II Pneumocytes

- Granular cells.
- Synthesize and secrete surfactant.
- Surfactant is primarily DPPC/lecithin. 🧪

💡 Type II cells can also proliferate and replace damaged Type I and Type II cells.

◆ Type III Pneumocytes — Alveolar Macrophages

- Large phagocytic cells.
- Maintain sterility by removing debris and microbes. 🛡️

💡 They provide an important immune defense within the alveoli.

🧪 Surface Tension and the Law of Laplace

The fluid lining the alveoli creates surface tension that tends to collapse the air sacs.

This is governed by the Law of Laplace:

 $P = 2T/r$

Where:

P = collapsing pressure

T = surface tension

r = radius

Without surfactant:

Smaller alveoli have a smaller r.

Therefore, they generate higher pressure P.

Smaller alveoli would collapse into larger ones. 🫁 ⬇️

💧 Surfactant

- Reduces surface tension.

- Stabilizes smaller alveoli.

- Increases lung distensibility.

A deficiency in surfactant causes Infant Respiratory Distress Syndrome (IRDS), leading to severe alveolar collapse.

💡 Why is surfactant important?

It prevents alveolar collapse, especially during expiration, when alveolar volume becomes smaller.

It also:

Decreases the work required to inflate the lungs.

Helps keep alveoli relatively uniform in size.

Reduces the tendency of fluid to enter the alveoli.

📊 Lung Compliance and Work Factors

Compliance = measure of lung expandability per unit change in transpulmonary pressure.

Compliance = $\Delta V / \Delta P$

📌 Normal value = 200 mL/cm H₂O

Elasticity is determined by two forces:

*1/3 → lung tissue fibers

*2/3 → surface tension

↑ Increased Compliance

Seen in:

*Emphysema

*Aging lung

→ Due to destruction of elastic tissue.

💡 High compliance means the lungs are easy to expand, but they may have difficulty recoiling during expiration.

↓ Decreased Compliance

Caused by:

*Pulmonary fibrosis

*Alveolar edema

*Increased pulmonary venous pressure (engorgement)

💡 Low compliance means the lungs are stiff and require greater pressure to produce the same change in volume.

Compliance: Important Clinical Comparison

Emphysema

Compliance ↑

➡ Lungs become easier to inflate but have poor elastic recoil.

Pulmonary Fibrosis

Compliance ↓

➡ Lungs become stiff and difficult to expand.

Lung Volume Relationships

Inspiratory Capacity

$$IC = TV + IRV$$

The maximum amount that can be inspired after a normal expiration.

Functional Residual Capacity

$$FRC = ERV + RV$$

The amount remaining in the lungs after a normal expiration.

💡 FRC acts as an important oxygen reservoir between breaths.

🫁 Vital Capacity

$$VC = IRV + TV + ERV$$

The maximum amount of air that can be moved voluntarily.

🫁 Total Lung Capacity

$$TLC = VC + RV$$

The total amount of air contained in the lungs after maximal inspiration.

🧪 Advanced Volume Measurement

RV, FRC, and TLC cannot be measured by direct spirometry.

The helium dilution technique is used, applying the mass conservation equation:

$$\text{🧮 } C_1 \times V_1 = C_2 \times (V_1 + V_2)$$

Where V_2 is the unknown FRC.

💡 Why can't spirometry measure RV?

Because after maximal expiration, the remaining air cannot be voluntarily exhaled into the spirometer.
Therefore, any capacity containing RV also cannot be measured directly by simple spirometry:

*RV ✗

*FRC ✗

*TLC ✗

Ventilation Efficiency

Anatomic Dead Space

Air in conducting passages:

➡ Approximately 150 mL

This air does not participate directly in gas exchange because it remains in the conducting zone.

Physiologic Dead Space

Anatomic dead space + non-functional or poorly perfused alveoli

 In a healthy person, physiologic dead space is approximately equal to anatomic dead space.

Alveolar Ventilation (VA)

Calculated as:

$$V_A = \text{Freq} \times (TV - VD)$$

At rest:

$$12 \times (500 - 150) = 4.2 \text{ L/min}$$

So although total minute ventilation is:

$$12 \times 500 = 6 \text{ L/min}$$

Only approximately 4.2 L/min reaches functioning gas-exchange areas.

📌 Strategically, increasing tidal volume is more effective than increasing respiratory rate because it reduces the proportion of each breath wasted in the dead space.

💡 Example

If breathing becomes very rapid and shallow:

TV ↓ → dead-space proportion ↑ → alveolar ventilation becomes less efficient.

If tidal volume increases appropriately:

TV ↑ → greater proportion reaches alveoli → alveolar ventilation ↑.

📺 Obstructive vs Restrictive Patterns

🗣️ Obstructive Disease

Examples:

*Asthma

*Emphysema

Main problem:

➡ Difficulty getting air OUT.

Typical finding:

FEV₁ ↓↓↓

 Restrictive Disease

Examples:

*Pulmonary fibrosis

*Severe chest-wall restriction

Main problem:

➡ Difficulty expanding the lungs.

Typical finding:

TLC ↓

 Easy memory:

Obstructive = Can't get air OUT 

Restrictive = Can't get air IN 🫁

3 🩸 PULMONARY CIRCULATION, CAPILLARY DYNAMICS & V/Q MATCHING

The lungs utilize a unique dual circulatory system to manage gas exchange and tissue nourishment.

🩸 Circulatory Comparison

● Pulmonary Circulation

- Low-pressure
- High-flow
- Supplies venous blood to alveolar capillaries.

💡 Why low pressure?

The pulmonary circulation receives the entire cardiac output, so its low resistance and low pressure protect the delicate pulmonary capillaries from excessive fluid filtration.

● Bronchial Circulation

- High-pressure
- Low-flow
- Systemic supply for the conducting airways.
- Represents 1–2% of cardiac output.

Bronchial veins from large airways return about half this flow via the azygos vein to the right heart.

The other half drains directly into the pulmonary veins, creating a 2% physiological shunt of deoxygenated blood into the oxygenated supply.

💡 High-yield:

Pulmonary circulation → gas exchange 🫁

Bronchial circulation → nourishes lung tissues 🩸

⚠️ Autoregulation and Hydrostatics

Pulmonary vessels exhibit Hypoxic Pulmonary Vasoconstriction.

When alveolar PO_2 falls below 73 mm Hg, adjacent vessels constrict to redirect blood to better-ventilated regions. 🔄 🫁

💡 Why is this useful?

If one area of the lung is poorly ventilated, sending large amounts of blood there would waste perfusion.

➡ Vasoconstriction redirects blood toward areas where ventilation is better.

✳️ This is different from most systemic tissues, where hypoxia generally causes vasodilation.

📌 West's Zones of Pulmonary Flow

Gravity creates hydrostatic pressure gradients in the upright lung.

These are defined by the relationships between:

- Alveolar pressure (P_A)
- Arterial pressure (P_a)
- Venous pressure (P_v)

Zone 1

$$P_A > P_a > P_v$$

 No flow.

- Alveolar pressure compresses capillaries.
- Rare in health.
- Occurs if P_a is low or P_A is high.

 Think:

Alveolar pressure wins → capillaries collapse → no blood flow.

Zone 2

$$P_a > P_A > P_v$$

- Intermittent flow during systolic peaks.
- Found in the lung apices.

💡 Blood flows when arterial pressure temporarily exceeds alveolar pressure.

■ Zone 3

$$P_a > P_v > P_A$$

- Continuous flow.

- Found in the lower lung regions.

💡 Here, both arterial and venous pressures exceed alveolar pressure, so the capillaries remain open.

🏃 During exercise, increased P_a converts Zone 1/2 regions into Zone 3 via capillary recruitment.

🧠 Easy memory:

Zone 1 → No flow ❌

Zone 2 → Intermittent flow 🔄

Zone 3 → Continuous flow ✅

💧 Starling Forces and Edema

Fluid exchange is governed by a balance of forces:

➡ Outward Forces = 29 mm Hg

- Capillary pressure = 7

- Interstitial colloid pressure = 14

- Negative interstitial pressure = 8

← Inward Forces = 28 mm Hg

- Plasma colloid osmotic pressure = 28

 Net Filtration Force

$29 - 28 = +1 \text{ mm Hg}$

This creates a slight fluid flow that is removed by lymphatics. 

 A small amount of fluid normally leaves the pulmonary capillaries, but the lymphatic system removes it, preventing pulmonary edema.

 Acute Safety Factor Against Edema

21 mm Hg

⚠ If capillary pressure exceeds 28 mm Hg (e.g., mitral disease) or membranes are damaged (e.g., chlorine gas), fluid fills the alveoli.

Pulmonary Edema

Excess fluid in the alveolar/interstitial spaces:

- ➔ increases diffusion distance
- ➔ reduces gas exchange efficiency
- ➔ can cause severe respiratory difficulty.

Ventilation-Perfusion (V/Q) Mechanics

The V/Q ratio describes the relationship between:

Ventilation = amount of air reaching alveoli

and

Perfusion = amount of blood reaching alveoli

↑ Lung Apex:

V/Q ratio ≈ 2.5

➔ Represents a dead-space-like condition.

There is relatively more ventilation compared with perfusion.

↓ Lung Base:

V/Q ratio ≈ 0.6

➔ Represents a shunt-like condition.

*There is relatively more perfusion compared with ventilation.

Both ventilation and perfusion increase from the top of the lung to the bottom.

However: Perfusion increases more rapidly than ventilation.

Therefore, the V/Q ratio is lowest at the base.

V/Q Mismatch

▲ High V/Q

There is:

Ventilation $\uparrow$ relative to perfusion $\downarrow$

➡ Air reaches alveoli but blood flow is insufficient.

This resembles dead space.

▼ Low V/Q

There is:

Perfusion $\uparrow$ relative to ventilation $\downarrow$

➡ Blood reaches alveoli that are poorly ventilated.

This resembles a shunt.

Easy memory:

Dead space = ventilation without enough blood  

Shunt = blood without enough ventilation 🩸 🫁